CEFET/RJ

Data Analytics
Experimental Lines (time series)

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Overview

- Workflows
- Experimental lines
- Data analytics

[1] E. Ogasawara, C. Paulino, L. Murta, C. Werner, and M. Mattoso, 2009, Experiment line: Software reuse in scientific workflows, In: *Scientific and Statistical Database Management*, p. 264–272
[2] A. Marinho, D. de Oliveira, E. Ogasawara, V. Silva, K. Ocaña, L. Murta, V. Braganholo, and M. Mattoso, 2017, Deriving scientific workflows from algebraic experiment lines: A practical approach, *Future Generation Computer Systems*, v. 68, p. 111–127.

Data preprocessing

- Direct
- Sliding windows

https://github.com/eogasawara/mylibrary/blob/master/examples/s3preprocess-test.R

Direct

```
x <- load_series("sin")
dir_test <- function(x) {
  y <- outliers.boxplot(x)
  tt <- ts_train_test(y, test_size=5)
}
dir_test(x)
```

Sliding windows

```
sw_test <- function(prec, x, size) {
  tt <- ts_train_test(x, test_size=size, sw_size=size)

  prec <- ts_setup(prec, tt$train)

  io <- ts_sw_project(tt$train)

  input <- ts_normalize(prec, io$input)
  output <- ts_normalize(prec, io$output, arguments=input$arguments)

  print(sprintf("%s i=[%.3f,%.3f] - o=[%.3f,%.3f]", class(prec)[1],
        min(input$x), max(input$x), min(output$x), max(output$x)))

  plot(output$x, main=class(prec)[1])

  input_org <- ts_denormalize(prec, input$x, arguments=input$arguments)
  output_org <- ts_denormalize(prec, output$x, arguments=input$arguments)

  print(sprintf("%s i=%.3f - o=%.3f", class(prec)[1],
        abs(mean(input_org$x-io$input)), abs(mean(output_org$x-io$output))))
}

sw_test(ts_gminmax(), x, 10)
```

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Model training

- Separating training and test data
- Setup the model with preprocessing
- Train the model
- Predict test data
- Test prediction

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Train and test (general function)

```
train_test <- function(x, model, train, test, steps_ahead) {
  model <- ts_train(model, train)

  io <- ts_sw_project(test)

  prediction <- ts_predict(model, io$input)

  model <- ts_test(model, test, steps_ahead = steps_ahead)

  print(sprintf("%s %.2f%%", class(model)[1], 100*model$test_smape))

  ts_plot_series(c(model$train_value, model$test_value), model$train_pred,
        model$test_pred, class(model)[1])

  return(model)
}
x <- load_series("sin")

ttx <- ts_train_test(x, test_size=5, sw_size=0)
train_test(x, model=ts_arima(), ttx$train, ttx$test, steps_ahead = 5)
```

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Train and test (direct data)

```
x <- load_series("sin")

 ttx <- ts_train_test(x, test_size=5, sw_size=0)
 train_test(x, model=ts_arima(), ttx$train,
      ttx$test, steps_ahead = 5)
 train_test(x, model=ts_eelm_dir(4), ttx$train,
      ttx$test, steps_ahead = 5)
 train_test(x, model=ts_emlp_dir(4), ttx$train,
      ttx$test, steps_ahead = 5)
```

Train and test (sliding window data)

```
x <- load_series("sin")
preprocess <- ts_gminmax()
 ttx <- ts_train_test(x, test_size=5, sw_size=10)
 train_test(x, model=ts_nnet(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_svm(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_rf(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_elm(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_mlp(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_tensor_cnn(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
 train_test(x, model=ts_tensor_lstm(preprocess, input_size=4),
      ttx$train, ttx$test, steps_ahead = 5)
```

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Training and testing

```
ts_train_test <- function(x, test_size, sw_size = 0, offset=0) {
  if (offset == 0) {
    offset <- length(x)-test_size
  }
  if (sw_size == 0) {
    train <- x[1:offset]
    test <- x[(offset+1):(offset+test_size)]
  }
  else {
    train <- x[1:offset]
    test <- x[(offset-(sw_size-1)+1):(offset+test_size)]
    train <- ts_sw(train, sw_size)
    test <- ts_sw(test, sw_size)
  }
  return(list(train=train, test=test))
}
```

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Preprocessing class

```
ts_preprocess <- function() {
  value <- list(sw_size = NA, scale = 1, offset = 0, rescale = FALSE)
  attr(value, "class") <- "ts_preprocess"
  return(value)
}

ts_setup <- function(obj, x) {
  #x contains both input and output
  UseMethod("ts_setup")
}

ts_normalize <- function(obj, x, arguments=NULL) {
  # x can be either input matrix or output vector
  UseMethod("ts_normalize")
}

ts_denormalize <- function(obj, x, arguments=NULL) {
  # x can be either input matrix or output vector
  UseMethod("ts_denormalize")
}
```

https://github.com/eogasawara/mylibrary/blob/master/examples/s3preprocess-test.R

Model class

```
ts_model <- function() {
  obj <- list(mdl=NULL)
  attr(obj, "class") <- "ts_model"
  return(obj)
}

ts_invoke_train <- function(obj, X, Y = NULL, arguments = NULL) {
  UseMethod("ts_invoke_train")
}

ts_train <- function(obj, x, arguments = NULL) {
  UseMethod("ts_train")
}

ts_invoke_predict <- function(obj, X) {
  UseMethod("ts_invoke_predict")
}

ts_predict <- function(obj, X, steps_ahead=1) {
  UseMethod("ts_predict")
}

ts_test <- function(obj, test, steps_ahead=1) {
  UseMethod("ts_test")
}
```

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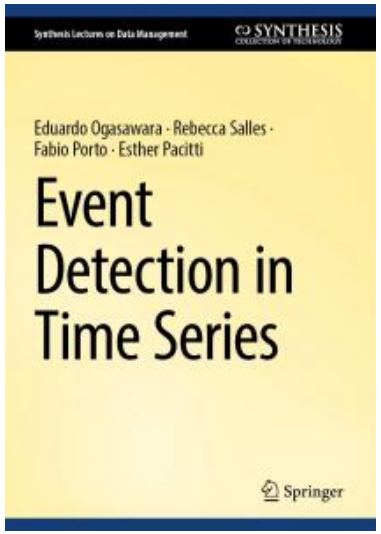
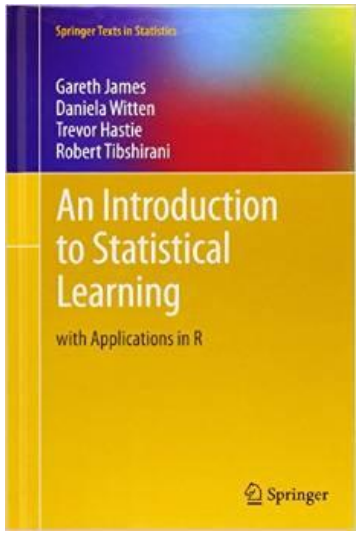
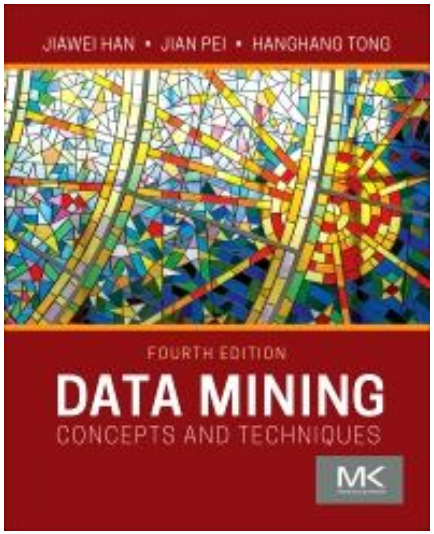
Practicing

- Take some time to practice the example
  - https://github.com/eogasawara/mylibrary/blob/master/examples/s3model-test.R
- Exercise
  - Run the examples using different preprocessing methods (adaptive normalization, swminmax)

Referências







[1] Ogasawara, E.; Salles, R.; Porto, F.; Pacitti, E. Event Detection in Time Series. 1. ed. Cham: Springer Nature Switzerland, 2025.

[2] Cryer, J. D.; Chan, K.-S. Time Series Analysis: With Applications in R. Springer Science & Business Media, 2008.

[3] Han, J.; Pei, J.; Tong, H. Data Mining: Concepts and Techniques. 4th edition ed. Cambridge, MA: Morgan Kaufmann, 2022.

[4] James, G. M.; Witten, D.; Hastie, T.; Tibshirani, R. An Introduction to Statistical Learning: With Applications in R. [s.l.] Springer Nature, 2021.